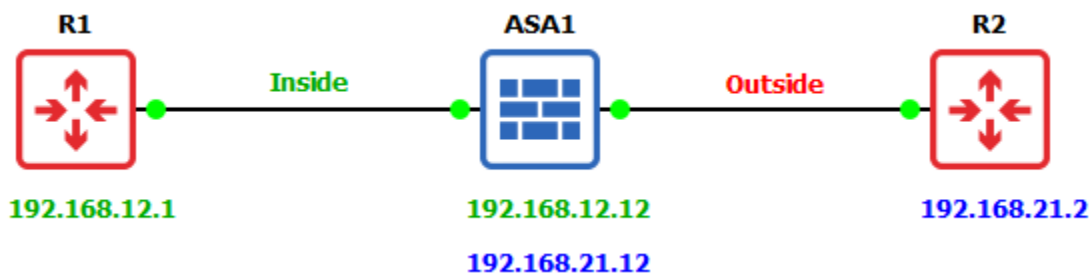


Routed Firewall & Transparent Firewall:

- o There are two modes in which you can have firewall; routed or transparent mode.
- o Each mode of ASA FW will treat the packets differently and operate in its own way.
- o Depending on your requirements of your design you will choose what is best for you.
- o Use case of Transparent Firewall to deploy between same subnet users and servers.
- o Another use case of Transparent firewall mode to deploy inside to connect with MPLS.

Routed Mode:

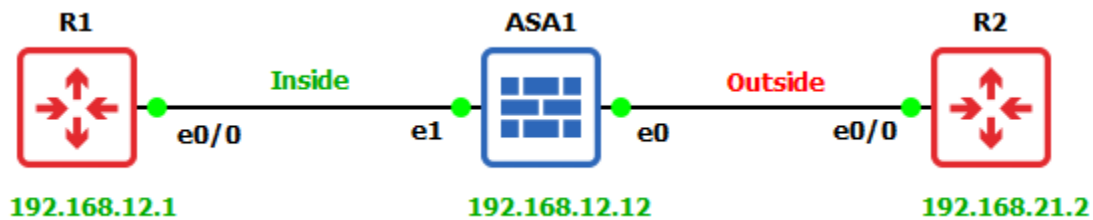
- o The Routed Firewall is the default mode for Cisco ASA Next Generation Firewall.
- o The Routed Firewall mode of ASA uses dynamic routing protocols & static routes.
- o Routed Mode supports multiple interfaces; each interface is on a different subnet.
- o The Routed Firewall Mode in each interface requires an IP address on that subnet.
- o Routed Mode perform Network Address Translation between connected networks.
- o Cisco ASA Firewall, Routed Mode is the most commonly use mode in the network.



Transparent Mode:

- o In Transparent Mode, the Cisco ASA Next Generation functions as a Layer 2 device.
- o The ASA operate in a secure bridging mode to provide rich OSI L2 through 7 security.
- o In this mode appliance is invisible to devices on both sides of a protected network.
- o ASA requires only one Management IP address assign to the Bridge Virtual Interface.
- o ASA Transparent mode does not support dynamic routing protocols, VPNs, QoS etc.
- o Cisco ASA, Transparent deployment allows easy integration with the existing network.
- o When change ASA firewall mode to transparent all existing configuration will erased.
- o ARP and BPDU is allowed from higher to lower and from lower to higher security level.
- o Use firewall transparent command to change the Cisco ASA FW to transparent mode.
- o Acts **bump in the wire** or **stealth** firewall not seen as router hop to connected devices.
- o It works at OSI layer 2, or it forwards the frames based on destination MAC Address.
- o But Transparent Firewall still has capabilities to filter the traffic from layer 2 to layer 7.
- o If you want to implement firewall in your network without readdressing the network.
- o Only 2 interface can use, no dynamic routing, no VPN only site-site vpn can configure.
- o In transparent Firewall no CDP, no DTP, no VTP no IPv6, no DHCP Relay Service available.

Lab Time:



R1 Configuration

```
R1(config)#interface Ethernet 0/0
R1(config-if)#ip address 192.168.12.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#mac-address aaaa.aaaa.1111
```

```
R1(config-if)#line vty 0 4
R1(config-line)#transport input all
R1(config-line)#password 123
R1(config-line)#login
R1(config-line)#exit
R1(config)#enable password 123
```

R2 Configuration

```
R2(config)#interface Ethernet 0/0
R2(config-if)#ip address 192.168.12.2 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#mac-address aaaa.aaaa.2222
```

```
R2(config-if)#line vty 0 4
R2(config-line)#transport input all
R2(config-line)#password 123
R2(config-line)#login
R2(config-line)#exit
R2(config)#enable password 123
```

Transparent Firewall Configuration
ASA1(config)# show firewall
ASA1(config)# firewall transparent
ASA1(config)# interface Ethernet 1
ASA1(config-if)# nameif inside
ASA1(config-if)# security-level 100
ASA1(config-if)# no shutdown
ASA1(config-if)# bridge-group 1
ASA1(config-if)# interface Ethernet 1
ASA1(config-if)# nameif outside
ASA1(config-if)# security-level 0
ASA1(config-if)# no shutdown
ASA1(config-if)# bridge-group 1
ASA1(config-if)# interface bVI 1
ASA1(config-if)# ip address 192.168.12.12 255.255.255.0
ASA Verification and Testing Commands
ASA1# show mac-address-table
ASA1# show mac-address-table interface inside
ASA1# show mac-address-table interface outside
ASA1# show run interface bVI 1
ASA1(config)# show mac-address-table
ASA1(config)# mac-address-table static inside aaaa.aaaa.1111
ASA1(config)# access-list 100 permit tcp any host 192.168.12.1 eq 23
ASA1(config)# access-group 100 in interface outside
R2#telnet 192.168.12.1
ASA1# show conn
ASA1(config)# no firewall transparent
ASA1# clear mac-address-table
Ethertype ACL Allow CDP Traffic
ASA1(config)# access-list test ethertype permit 0x2000
ASA1(config)# access-group test in interface outside
ASA1(config)# access-group test in interface inside
Verification
R1#show cdp neighbors
R2#show cdp neighbors